



SECP2523: DATABASE

SEM I 2024/2025

DATABASE DESIGN DESCRIPTIONS: CONCEPTUAL DATABASE DESIGN FOR KADA Cooperative System

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Stakeholder: Lembaga Kemajuan Pertanian Kemubu (KADA)

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NOTE: A module can consist of one or multiple (related) use case. A team member in a group must be responsible to at least one module.

1. Introduction

1.1 Overview about the company

KADA (Lembaga Kemajuan Pertanian Kemubu) is an agriculture organization in Kelantan. The workers under KADA are being given the opportunity to become cooperative members of KADA and make a loan under KADA too. But, the current system, or the as-is system in KADA to provide such services is being done manually and there is where we saw that it has very low efficiency and productivity. According to the list below, are the services that KADA provided to their staff currently.

1. Registration to become a membership of cooperative of KADA

In the current system, all the registration procedures have to undergo a manual process, such as filling in the paper form. For example, users must pick up the form in person at the headquarters in order to fill it in. In the registration process, most of the personal information of the users that are private and sensitive will be collected. Then, the users have to submit it manually also by submitting it at the headquarters. After the filled in and submitting process is done, then the registration paper would be further reviewed by the staff itself manually whether the users are qualified to become their member or not.

2. Applying a loan in cooperative of the KADA

Same as the registration process to become a membership of KADA, applying for a loan is also by filling in the paper form manually. The difference is in applying for a loan form, it would ask more detail about the user income information and guarantor. After the users have submitted it into the headquarters, it would be reviewed by the staff and the approval would be given by the higher position. The approval is given by conducting a monthly meeting between the staff and the higher position.

The current system has a lot of disadvantages, that is why we proposed a solution by making all these services online. By having it online, it could save a lot of time for the users and KADA, and it could also increase the KADA work efficiency.

1.2 Problems

Based on our analysis, we've identified several issues in the current system that contribute to its inefficiency and reduce its effectiveness in performing tasks smoothly and efficiently.

1. Loss of information

Most of the records are stored in paper such as the registration documents and also the loan documents. Paper documents can be easily damaged by water, fire or mishandling. Once damaged, the information on them may become unreadable or entirely lost. Other than that, paper documents can be misplaced and lost due to human error.

2. Accumulation of data when access or update

The clerk must locate the paper back in order to update the data, which is a difficult operation because they must scan and identify each paper in their safe in order to locate the particular one. As a result, it will take longer to finish a single task—updating the user's information.

3. Highly dependent on human

Using paper forms greatly reduces productivity and efficiency because handling, filing, and finding these documents take a lot of time and effort. The dependence on paper slows down operations, causes delays, and keeps employees from focusing on more important tasks, ultimately making the organization less productive.

4. Difficulty in distributing and receiving the application

In order to obtain the form and submit it, applicants must make an extensive journey to the headquarters. They are certain to be burdened by this since they will need to find time to travel to the headquarters and spend money on transportation.

1.3 Proposal

The system will serve as a cutting-edge substitute for the outdated paperwork system. Thus, the system will perform the same basic tasks as the original paperwork system, including handling the registration form for the new member and also the registration form to apply for a loan for the member of the organization. On the other hand, The shift from physical documentation to an online form is the main upgrade for the system. In the past, the organization primarily employed paper in their operations to store user information and to verify users during the loan application and registration processes. Initially, the current system only had three primary user views: the applicator, Ahli Lembaga Pengarah (ALK), and the clerk. However, after a few discussions, we concluded that the admin roles would be essential for the new online system.

Firstly, from the view of the applicants, The new system will definitely be beneficial for the user as the online system offers many benefits for the applicants. With the current paper system, people have to travel to the headquarters to get and submit forms, which takes a lot of time and money. This can especially be difficult for those who live far away or have busy schedules. In contrast, the online system allows people to access and submit forms anytime and from anywhere with an internet connection. This means they don't need to make a special trip, saving both time and travel costs. The online system is available 24/7, making it much more convenient. It also provides instant confirmation when an application is submitted, so applicants don't have to worry whether their forms are received by KADA or not. Additionally, online forms often help prevent mistakes by checking for missing or incorrect information before submission. Overall, the online application system makes the process faster, easier, and cheaper for everyone involved.

Secondly, for the members of the Board of Directors (ALK), their job of checking membership and loan applications becomes much easier with the new online system. Before, this process was slow and difficult because they had to be at the office and deal with lots of paperwork, which limited their flexibility and caused delayed decisions. With the online system, ALK members can do their work from anywhere and at any time, without needing to travel or handle physical documents. This means they can approve applications faster and more easily. The system also provides instant updates and notifications, making their work more efficient. As a result, the organization becomes much more productive in completing tasks that used to take days or weeks in a much shorter time. This makes the work of ALK members easier and improves the overall efficiency and speed of the organization.

The discussion revolves around creating a user-friendly system interface for clerks to handle applications smoothly. This interface allows clerks to easily access applicant details and payment records, and to update application statuses promptly. A key task for clerks is to review submitted forms, ensuring they meet the requirements set by KADA. With the system's filtering features, clerks can quickly sort through applications, accepting those that meet criteria and rejecting incomplete or non-compliant ones. By diligently updating database

records, clerks maintain accurate information for organizational use. This streamlined process helps clerks efficiently manage applications, contributing to the organization's overall effectiveness.

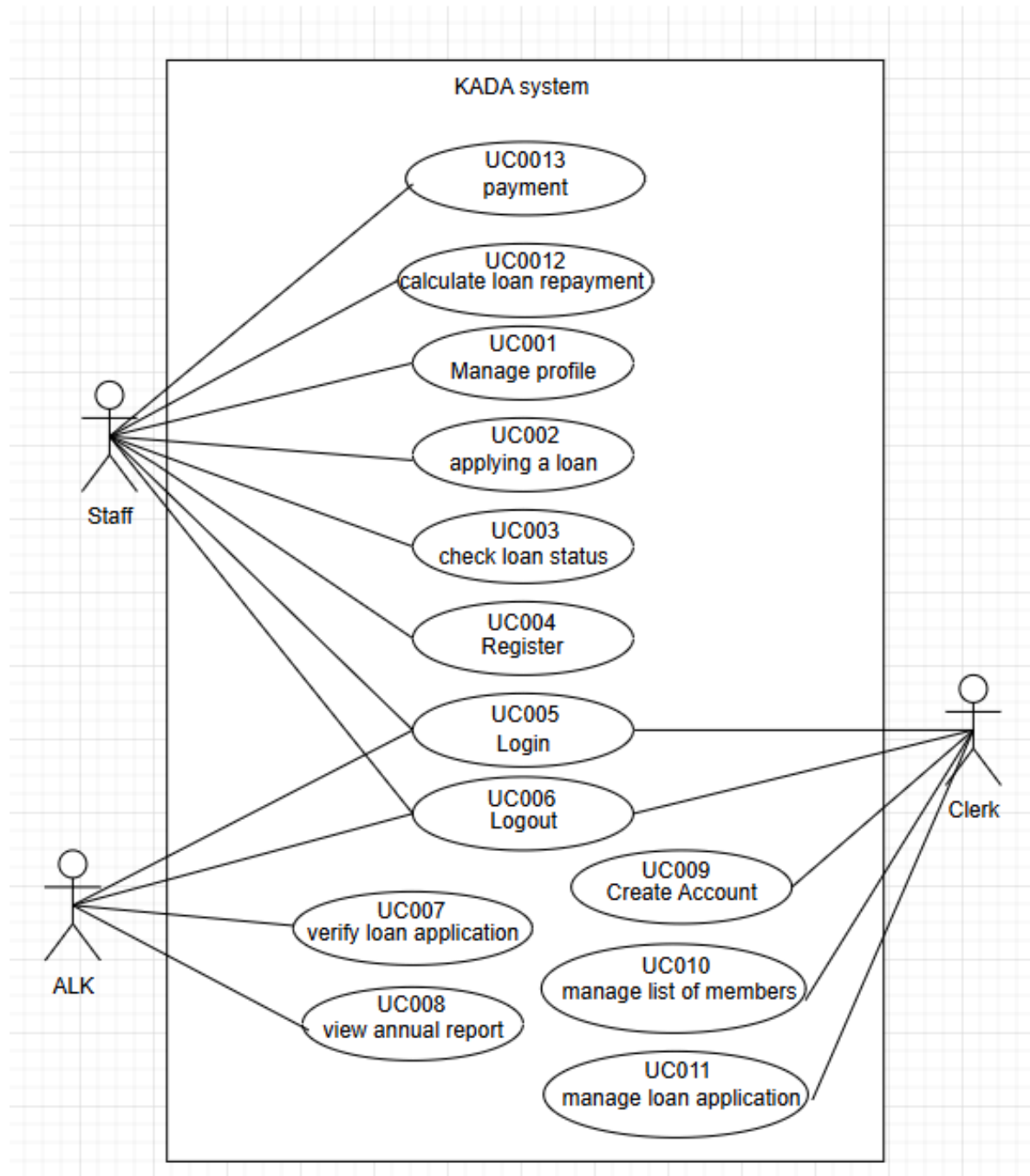
Lastly, for the administrative, their role in this new proposed system is to maintain the data and also the system itself. The administration needs to ensure that the system can run smoothly without any problem. Other than that, they also need to ensure that every application information is recorded in the database so that the data is safely secure.

1.4 Definitions, Acronyms and Abbreviations

- 1) KADA - Lembaga Pertanian Kemajuan Kemubu
- 2) ALK - Ahli Lembaga Pengarah

2. Database System Requirements

2.1 System overview



2.1 Module 1 <User>

Person in charge of this module is Syarifah Dania Binti Syed Abu Bakar

2.1.1 Module Description

This module focuses on managing user interactions within the cooperative system. It simplifies tasks like creating accounts, logging in, updating profiles, and securely logging out. By streamlining these operations, it ensures that members can access and manage their accounts with ease. The module includes four main functions:

1. **Manage Profile (UC001):** Allows members to view and update their personal details, ensuring their information stays current.
2. **Login (UC004):** Verifies user credentials and grants secure access to the system.
3. **Register (UC005):** Enables new members to create accounts, paving the way for their cooperative activities.
4. **Logout (UC006):** Ends user sessions securely, safeguarding account information.

2.1.2 Transaction Requirements: Data Entry

1. **Manage Profile (UC001):**
 - Members can edit their personal details, such as name, address, and contact information.
 - The system checks the validity of the input before saving changes.
2. **Register (UC005):**
 - Users provide their username and email, as supplied by the clerk.
 - The system processes the details and stores them to create a new account.

2.1.3 Transaction Requirements: Data Update/Deletion

1. **Manage Profile (UC001):**
 - Members can update their information, like changing their address or contact number.
 - Updates are only saved if all fields pass validation checks.

2. Register (UC005):

- If any required details are missing or incorrect, the system prompts the user to fix them and try again.

2.1.4 Transaction Requirements: Data Queries

1. Manage Profile (UC001):

- The system retrieves and displays current user details from the database for editing.
- Once updated, the changes are reflected in real-time.

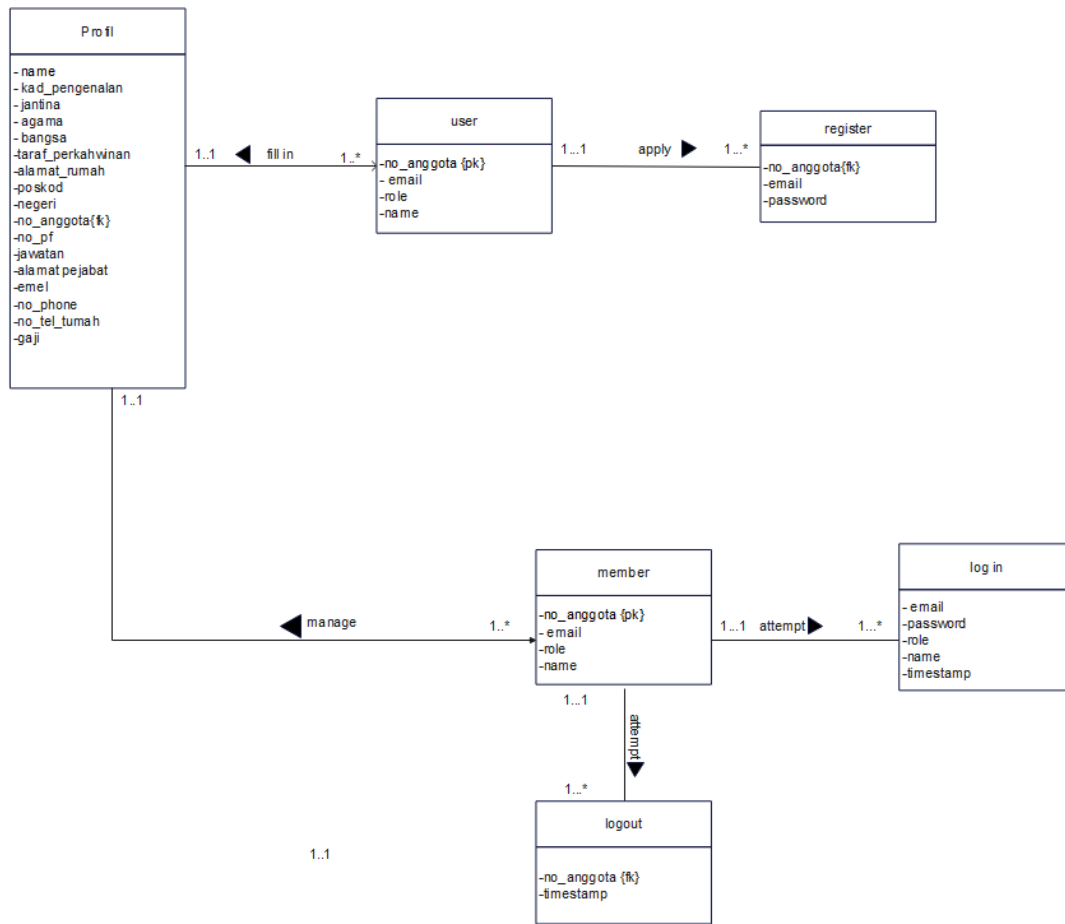
2. Login (UC004):

- The system validates login credentials against stored data.
- On successful login, it fetches user-specific information to generate the dashboard.

3. Logout (UC006):

- The system marks the user's session as inactive, ensuring secure termination of access.

2.1.5 Local Conceptual Data Model (ERD)



2.2. Module 2 <Loan Application>

The person in charge of this module is Yasmin Batrisyia Binti Zahiruddin

2.2.1. Module Description

This module will involve members of KADA only. There are 3 use case inside this module: UC002: applying a loan, UC003: check loan status, and UC0012: calculate loan repayment

UC002 - Applying loan

- Members who want to apply for a loan must be registered as a member of KADA in the system.
- Members could access the loan application form from the home page
- Members have to fill in all the requirements for the loan application.
- If the requirements are all correct, then it will be sent to the clerk, and ready for the approval phase from the ALK.

UC003 - Check loan status

- After members have submitted the loan application form, they are able to check the loan status in their profile, whether it is successful, in progress, pending, or unsuccessful.

UC0012 - Calculate loan repayment

- Members can get a rough view of how the repayment will be conducted later on.

UC0013 - Payment

-Members can make payments for their loans through the system. The payment process involves the following steps:

2.2.2. Transaction Requirement: Data Entry

UC002 - Applying loan

- Enters all the personal requirement details correctly by following the loan application form instruction
- Data from the loan application form will be entered into the database

UC0012 - Calculate loan repayment

- Enters the loan type, loan amount, and the duration of repayment for the calculation

UC0013 - Payment

Enters the payment details, including the payment amount and method. The system captures this data and processes the payment.

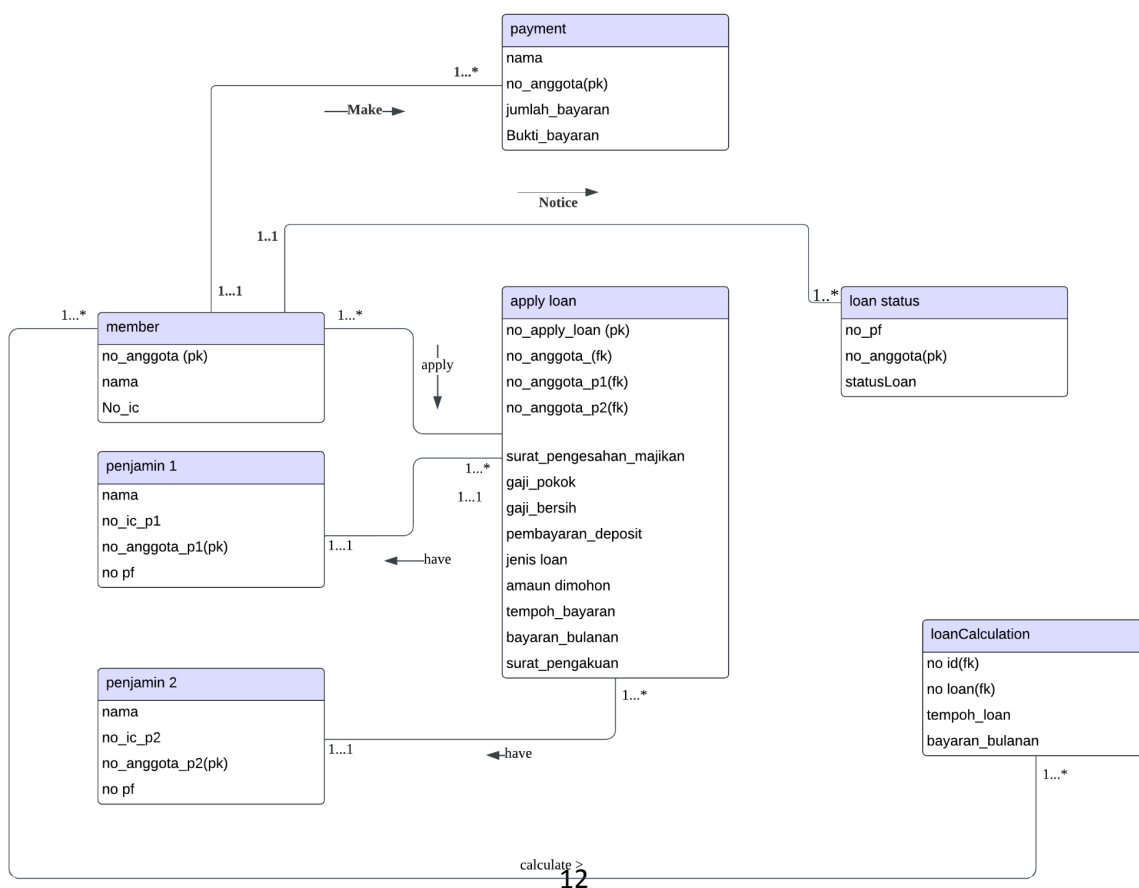
2.2.3 Transaction Requirement: Data Update/Deletion

1. If the requirements details are incorrect during the loan application phase, the system will ask the user to resubmit their application..

2.2.4. Transaction Requirement: Data Queries

1. The system will queries and extract the user details requirement from the loan application form to the database.

2.2.5 Local Conceptual Data Model (ERD) for Module 2



2.3. Module 3 <Clerk>

The person in charge of this module is Muhammad Naim Bin Abdullah

2.3.1 Module Overview

This module focuses solely on the clerk's responsibilities within KADA. It encompasses three use cases:

- **UC009:** Create an account
- **UC010:** Manage the list of members
- **UC011:** Manage loan applications

Detailed Functionality:

1. Creating a Member Account (UC009):

- The clerk is responsible for creating accounts for new members.
- To ensure the applicant is a KADA staff member, the clerk will verify the staff ID.

2. Managing Member Information (UC010):

- The clerk can monitor and update member information.
- If a member chooses to leave KADA or has retired, the clerk can delete their information from the system.

3. Managing Loan Applications (UC011):

- The clerk can review the details of loan applications.
- They are authorized to add remarks or comments explaining why a loan application should be approved or declined.
- The clerk can also view information about approved loan applications.
- If a loan application has been fully paid or requires deletion due to other circumstances, the clerk can delete it from the system.

2.3.2 Transaction Requirement: Data Entry

UC009 - Creating a Member Account

- input the username based on staff ID and password based on IC number

UC011 - Managing Loan Application

- make a remark or a comment to the Loan Application for ALK use

2.3.3 Transaction Requirement: Data Update/Deletion

UC009 - Creating a Member Account

- The clerk will deny the application if the new applicant is not connected to KADA.

UC010 - Managing Member Information

- If a member decides to leave the KADA cooperative or has retired, the clerk can delete their information from the database.

UC011 - Managing Loan Application

- If a loan application has been fully paid or requires removal due to other valid circumstances, the clerk is authorized to delete it from the system.

2.3.4. Transaction Requirement: Data Queries

UC009 - Creating a Member Account

- The system will queries and extract the member details that are on status “accepted” to the creating account form from the new member database.

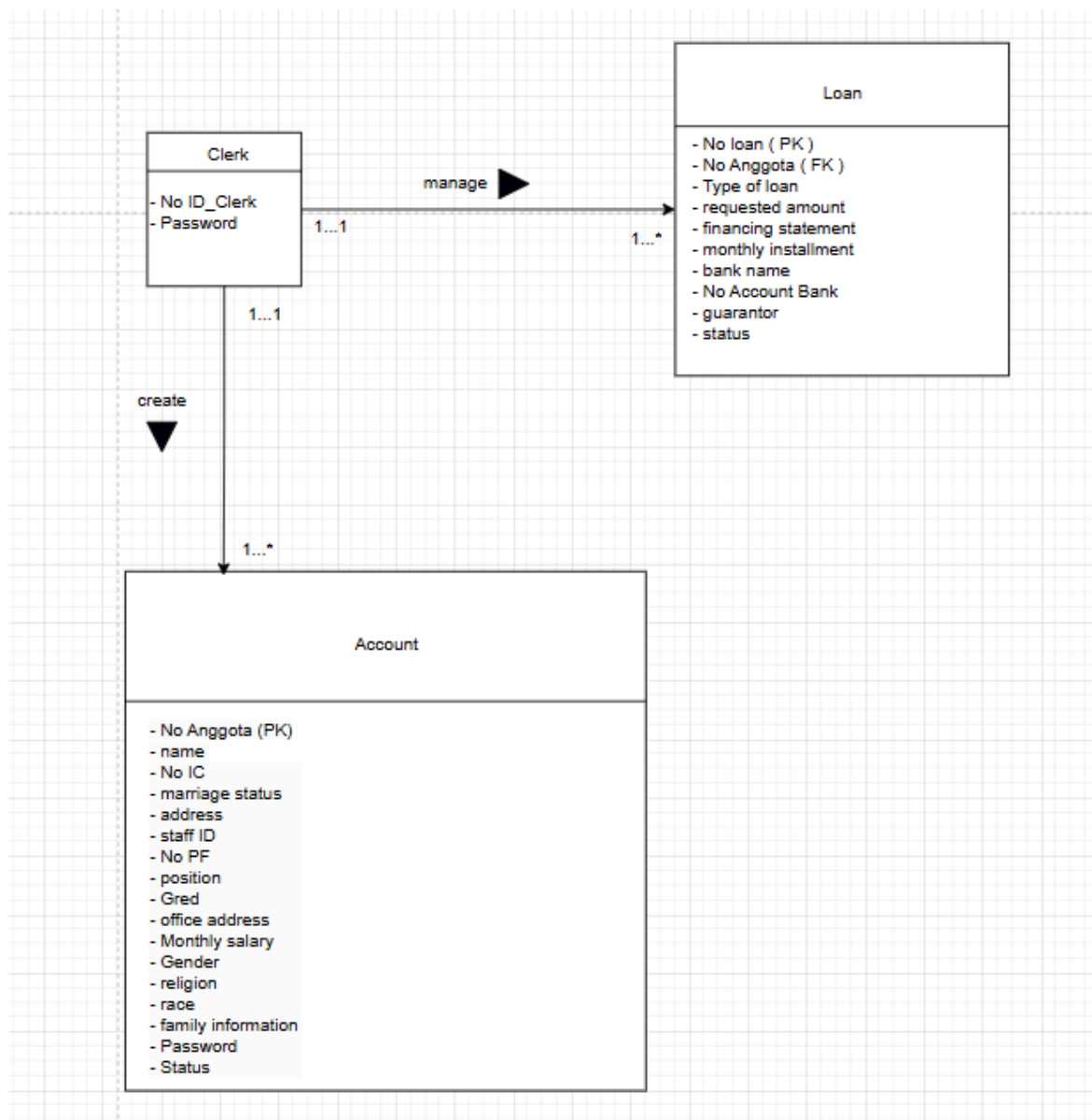
UC010 - Managing Member Information

- The system will queries and extract the member details from the new member form to the database.

UC011 - Managing Loan Application

- The system will queries and extract the loan application details from the new member form to the database.

2.3.5 Local Conceptual Data Model (ERD) for Module 3



2.4 Module 4 <ALK>

The person in charge of this module is Mukhritz.

2.4.1 Module Overview This module focuses on the ALK's responsibilities for verifying loan applications and viewing annual reports within the KADA system. It combines two use cases:

- UC007: Verify Loan Application
- UC008: View Annual Report

Detailed Functionality:

UC007 - Verify Loan Application

- ALK members log into the system to access the "Loan Applications" section.
- ALK selects a loan application from the available list to review its details.
- After reviewing, ALK can approve or reject the loan application.
 - Approval or rejection updates the loan application status in the system.
 - If the application has missing or invalid details, ALK can reject it with appropriate remarks.

UC008 - View Annual Report

- ALK members log into the system to access the "Reports" section.
- ALK selects the "Annual Report" from the available reports.
- The system displays a static annual report containing:
 - Financial summaries
 - Profit and loss data
 - Loan statistics
- ALK can review the report to analyze the cooperative's financial performance.

2.4.2 Transaction Requirement: Data Entry **UC007 - Verify Loan Application**

- ALK inputs remarks or comments for rejected loan applications to provide a justification.

2.4.3 Transaction Requirement: Data Update/Deletion **UC007 - Verify Loan Application**

- The system updates the loan application status to "Approved" or "Rejected" based on ALK's decision.
- If rejected, the application status will include the remarks provided by ALK.

2.4.4 Transaction Requirement: Data Queries **UC007 - Verify Loan Application**

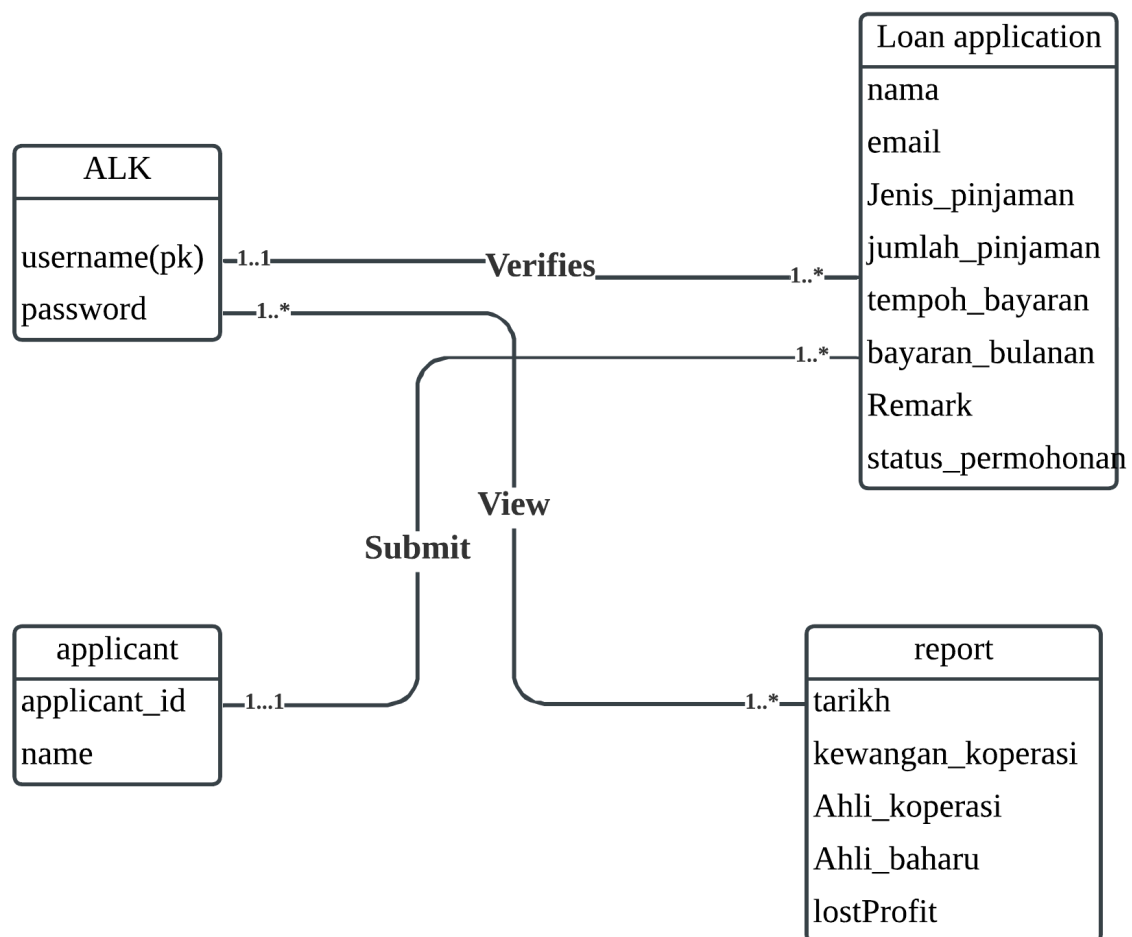
- The system queries and displays loan application details from the database for ALK's review.

UC008 - View Annual Report

- The system queries and retrieves the annual report data, including financial summaries and loan statistics, to present to ALK for analysis.

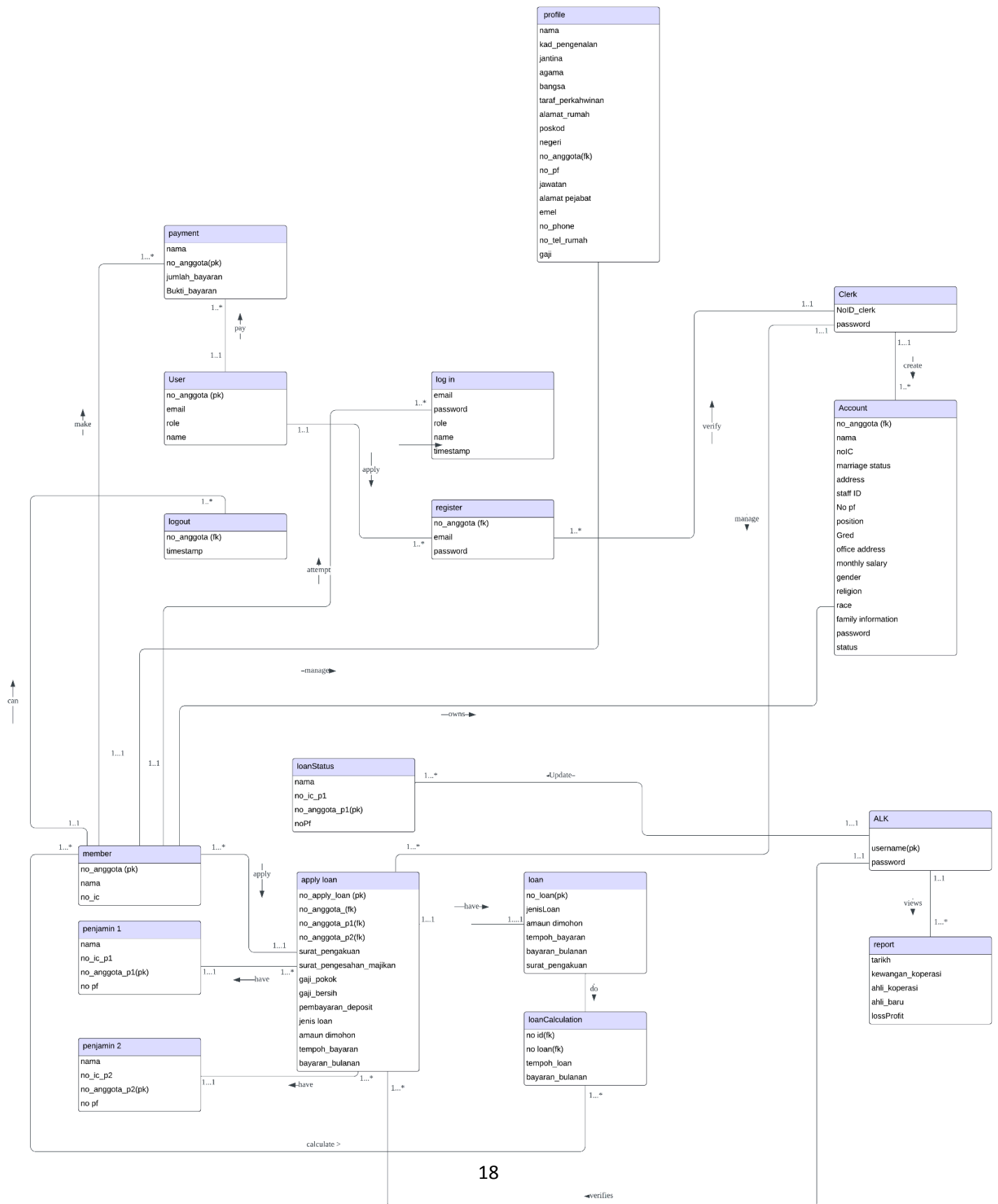
This module ensures efficient management of loan applications and seamless access to financial performance reports by ALK members, enhancing decision-making and operational transparency within KADA.

2.4.5 Local Conceptual Data Model (ERD) for

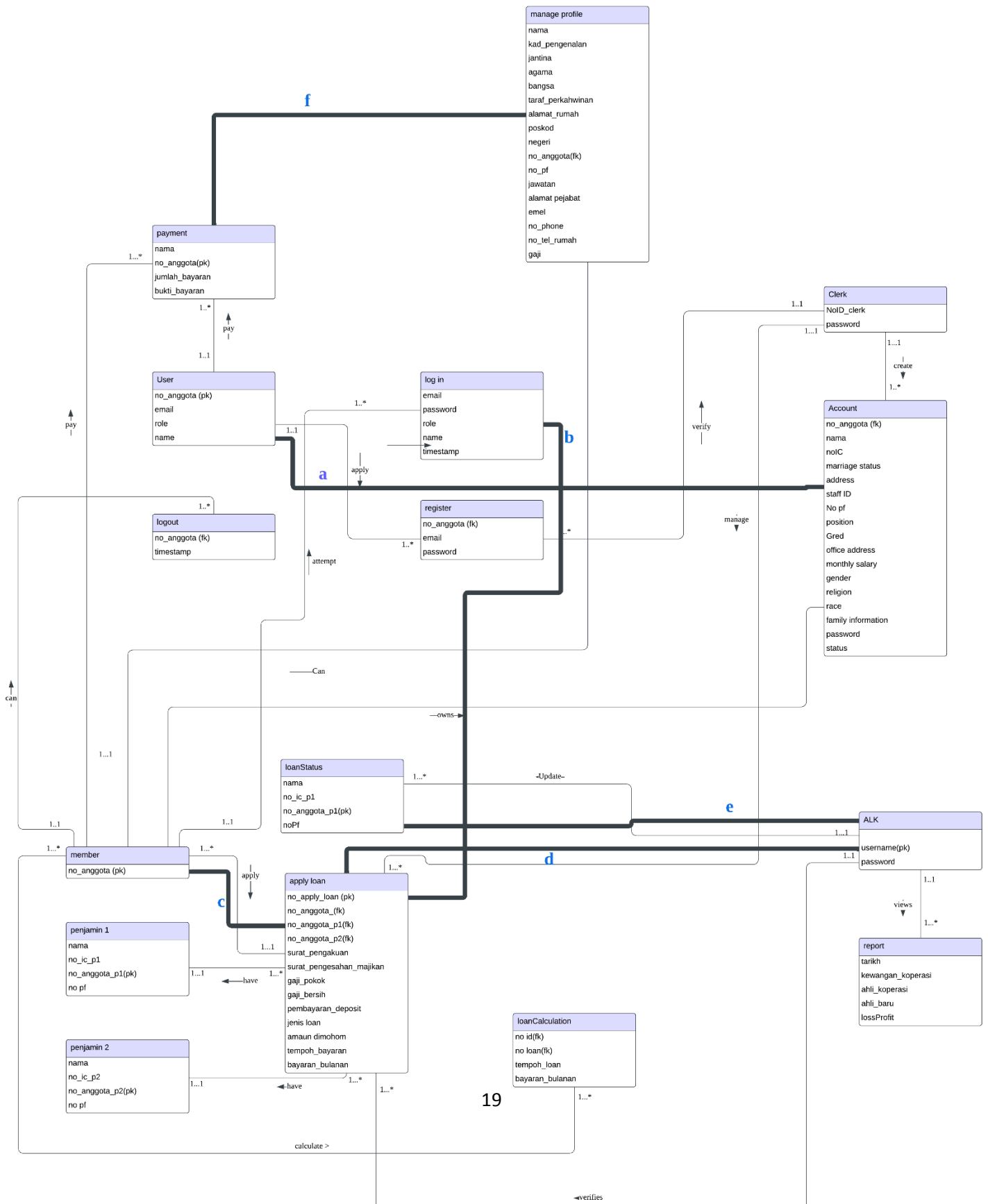


3. Global Conceptual Data Model (Global ERD)

3.1 Global Conceptual Design (ERD)



3.2 Verified Global Conceptual Design (Verified ERD)



Symbol	Transaction
a	2.1.2.1, 2.1.3.1, 2.1.4.1
b	2.1.4.1, 2.2.2.1, 2.2.3.1
c	2.1.4.1, 2.2.2.1, 2.2.3.1, 2.2.2.2
d	2.4.2.1, 2.4.4.1, 2.4.4.2
e	2.4.3.1, 2.4.3.2
f	2.2.3.1, 2.2.4.1

3.3 Data Dictionary for the global conceptual ERD

3.3.1 Identifying Entity

Entiti Name	Description	Aliases	Occurrence
member	User who already owns an account	User	occurs when a member has already log in
penjamin 1	Guarantor 1 is to verified the details accuracy for member in the loan application	Guarantor	occurs when member filled in for loan application
penjamin 2	Guarantor 2 is to verified the details accuracy for member in the loan application	Guarantor	occurs when member filled in for loan application
applyLoan	Loan application form for members	loan form	occurs when member want to make a loan from KADA
loanCalculation	give a rough view amount about the loan repayment calculation	loan calculator	occurs when members want to know how the loan repayment works before applying it
loanStatus	give a current loan status to the member whether it has been approved, pending or not approved	current loan state	occurs when member click on the profile
Clerk	A user who manages the loan application and membership	Admin	occur when the user login as a clerk
Account	details of the applicant for membership in the KADA	member information	occur when the clerk created an account for the member

ALK	A user who verify the loan application and view the report	executives	occur when user login as ALK
report	give a overall report about the cooperation	summary	occur when ALK click on the report
payment	give a user service to make a loan payment	transaction	occur when member want to make a payment

3.3.2 Identifying Relationship Types

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
User	1...* 1...*	Apply pay	Register payment	1...1 1...1
member	1...* 1..1 1...1 1...1 1...* 1...*	Apply Edit Attempt Own Pay Calculate	Apply Loan Profile Login Account Payment LoanCalculation	1...* 1...* 1...* 1...1 1...1 1...*
Clerk	1...* 1...* 1...* 1...*	Manage Create Verify Check	Apply Loan Account Register Apply Loan	1...1 1...1 1...1 1...1
ALK	1...* 1...* 1...*	Update Verify view	Loan Status Apply Loan Report	1...1 1...1 1...*

3.3.3 Attributes Description

Entity Name	Attributes	Data Type & length	Description	Nul l	Mul ti- valu ed	Example
member	no_anggota (PK)	Varchar (255)	Uniquely identify each membership number	No	No	A234
	nama	Varchar (255)	Member's full name	No	No	Nur Ain Binti Azim
	no_IC	Varchar (20)	Member identification card number	No	No	981223-10-3246
	email	Varchar (255)	Member's unique email	No	No	naimboekk@gmail.com

			address			
	role	Varchar (50)	Member's position in KADA	No	No	Clerk
penjamin 1	nama	Varchar (255)	Guarantor's 1 full name	No	No	Aina Husna Binti Ahmad
	no_ic_p1	Varchar (20)	Guarantor's 1 identification card number	No	No	991025-14-23 42
	no_anggota_p1 (PK)	Varchar (255)	Guarantor's 1 unique membership number	No	No	A541
	no_pf1	Varchar (50)	Guarantor's 1 registration number	No	No	RG2300
penjamin 2	nama	Varchar (255)	Guarantor's 2 full name	No	No	Nisa Sofea Binti Ali
	no_ic_p2	Varchar (20)	Guarantor's 2 identification card number	No	No	990822-14-24 28
	no_anggota_p2 (PK)	Varchar (255)	Guarantor's 2 unique membership number	No	No	A539
	no_pf2	Varchar (50)	Guarantor's 2 registration number	No	No	RG2500
applyLoan	no_apply_loan (PK)	int (10)	after filled in all the details, the application will be provided with the loan application serial number	No	No	L11234

	no_anggota (FK)	Varchar (255)	unique membership number by member	No	No	A234
	no_anggota _p1 (FK)	Varchar (255)	Guarantor 1 unique membership number by penjamin 1	No	No	A541
	no_anggota _p2 (FK)	Varchar (255)	Guarantor 2 unique membership number by penjamin2	No	No	A539
	jenis_loan	Varchar (255)	Type of loan wants apply	No	No	Al-Bai
	amaun_dim ohon	decimal (10,2)	Amount to borrowed from KADA	No	No	100001
	tempoh_ba yaran	int (10)	Duration of the loan repayment	No	No	10
	bayaran_bu lanan	decimal (10,2)	Monthly amount needs to be paid by the member	No	No	200
	surat_peng akuan	Varchar (255)	members upload the declaration letter files	No	No	SuratPengaku an.pdf
	surat_peng esahan_ma jikan	Varchar (255)	members upload the employer verification letter	No	No	suratPengesa hanMajikan.p df
	gaji_pokok	decimal (10,2)	basic salary of member	No	No	10000
	gaji_bersih	decimal	net salary of	No	No	7000

		(10,2)	member			
	pembayaran_deposit	Varchar (255)	the deposit transaction proof files by the member	No	No	deposit.pdf
loanStatus	no_anggota (FK)	Varchar (255)	unique membership number by member	No	No	A234
	no_pf	Varchar (255)	member registration number	No	No	RG1188
	statusLoan	text	loan current state	No	No	Approved
loanCalculation	no_anggota (FK)	Varchar (255)	unique membership number by member	No	No	A234
	no_loan (FK)	Varchar (10)	Unique loan number by loan	No	No	1123
	tempoh_loan	int (10)	Duration of the loan repayment	No	No	10
	bayaran_bulanan	decimal (10,2)	Monthly amount needs to be paid by the member	No	No	200
Clerk	email	varchar(255)	Account email for login	No	No	weq123@gmail.com
	password	varchar(255)	Account password for login	No	No	weq123@w

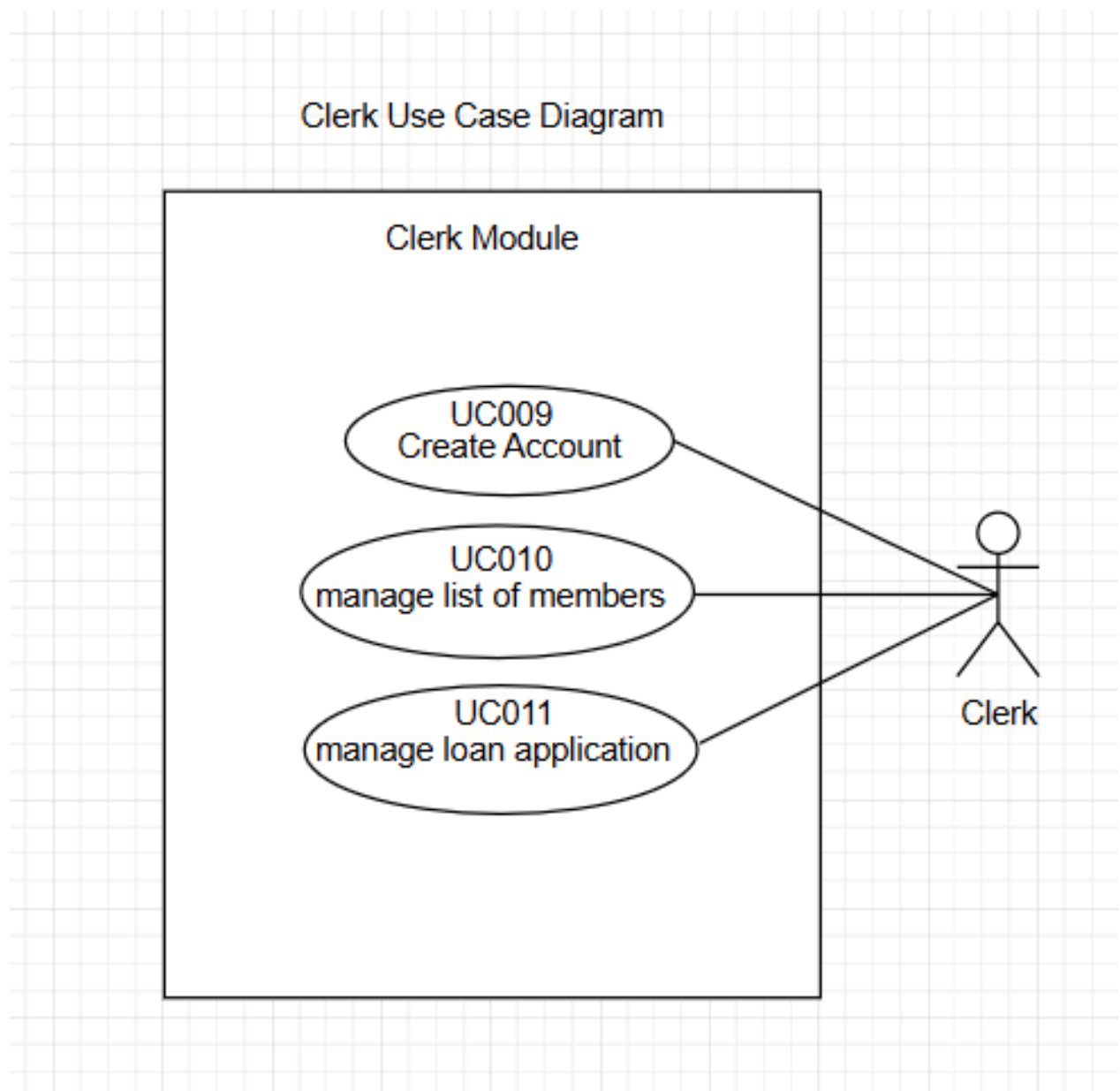
Account	No Anggota	varchar(255)	unique identifier for each member	No	No	A001
	Name	varchar(255)	member full name	No	No	Abu Bin Ahmad
	No IC	VARCHAR(20)	unique identifier for each member	No	No	123456-78-9012
	marriage status	varchar(255)	member's marriage status	No	No	single
	address	varchar(255)	member's home address	No	No	Universiti Teknologi Malaysia Johor Bahru 81310 UTM Johor Bahru, Johor, Malaysia.
	No PF	VARCHAR(50)	unique identifier for each member	No	No	B001
	position and Gred	VARCHAR(100)	the position and the gred of the member	No	No	officer GRED 41
	Office address	VARCHAR(100)	member's office address	No	No	Universiti Teknologi Malaysia Johor Bahru 81310 UTM Johor Bahru, Johor, Malaysia.
	monthly salary	INT	member's salary	No	No	2000
	Gender	VARCHAR(50)	member's gender	No	No	Male
	religion	VARCHAR	member's	No	No	Islam

		(50)	religion			
	race	VARCHAR (50)	member's race	No	No	Melayu
	password	varchar(2 55)	a unique password for the account	No	No	weq123@w
	Status	VARCHAR (50)	status of the member application	No	No	accepted
ALK	username	varchar (255)	ALK username	no	no	mukhriz Al iman
	password	varchar(2 55)	a uniques password for the account	no	no	123456abn.
report	tarikh	date	specific date for the report	no	no	12.3.2024
	kewangan_ koperasi	int	cooperative cash flow	no	no	150000
	ahli_kopera si	int	number of the cooperative member	no	no	250
	ahli_baharu	int	number of new member of the cooperative	no	no	20
	lossProfit	int	number of profit or loss that cooperative had	no	no	15000
payment	nama	varchar(2 55)	user full name	no	no	A001
	no_anggota	varchar(2 55)	unique identifier for each member	no	no	ahmad Adib
	jumlah_bay	int	payment make	no	no	2000

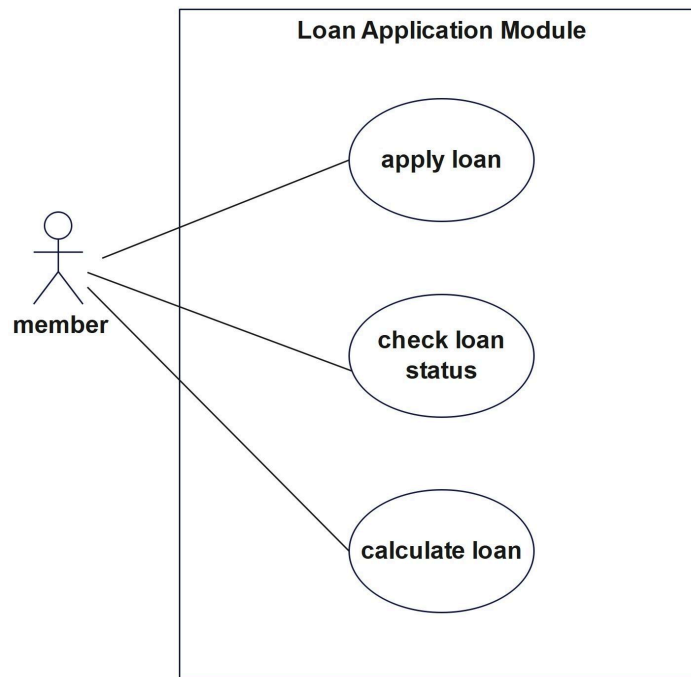
	aran		to the system			
	bukti_bayaran	Varchar (255)	user upload the transaction receipt	no	no	receipt.pdf

4. Appendices

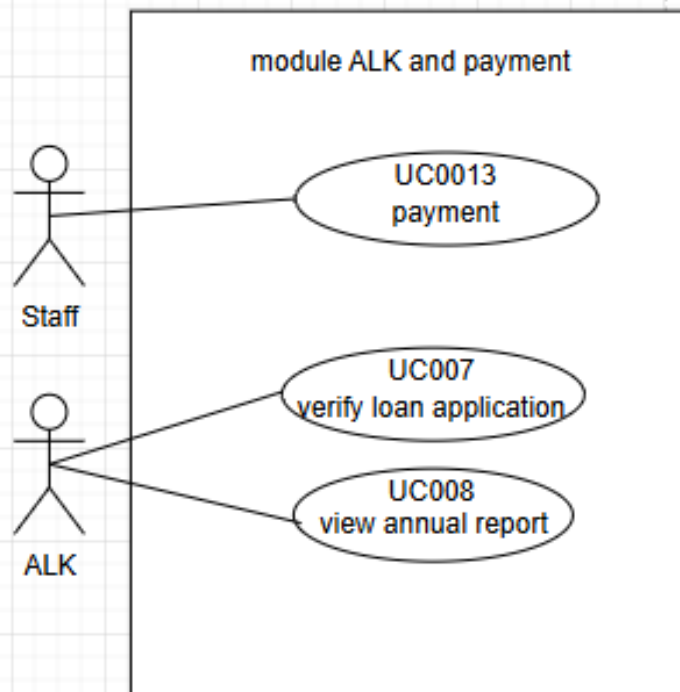
4.1 Use Case Diagram for the system.

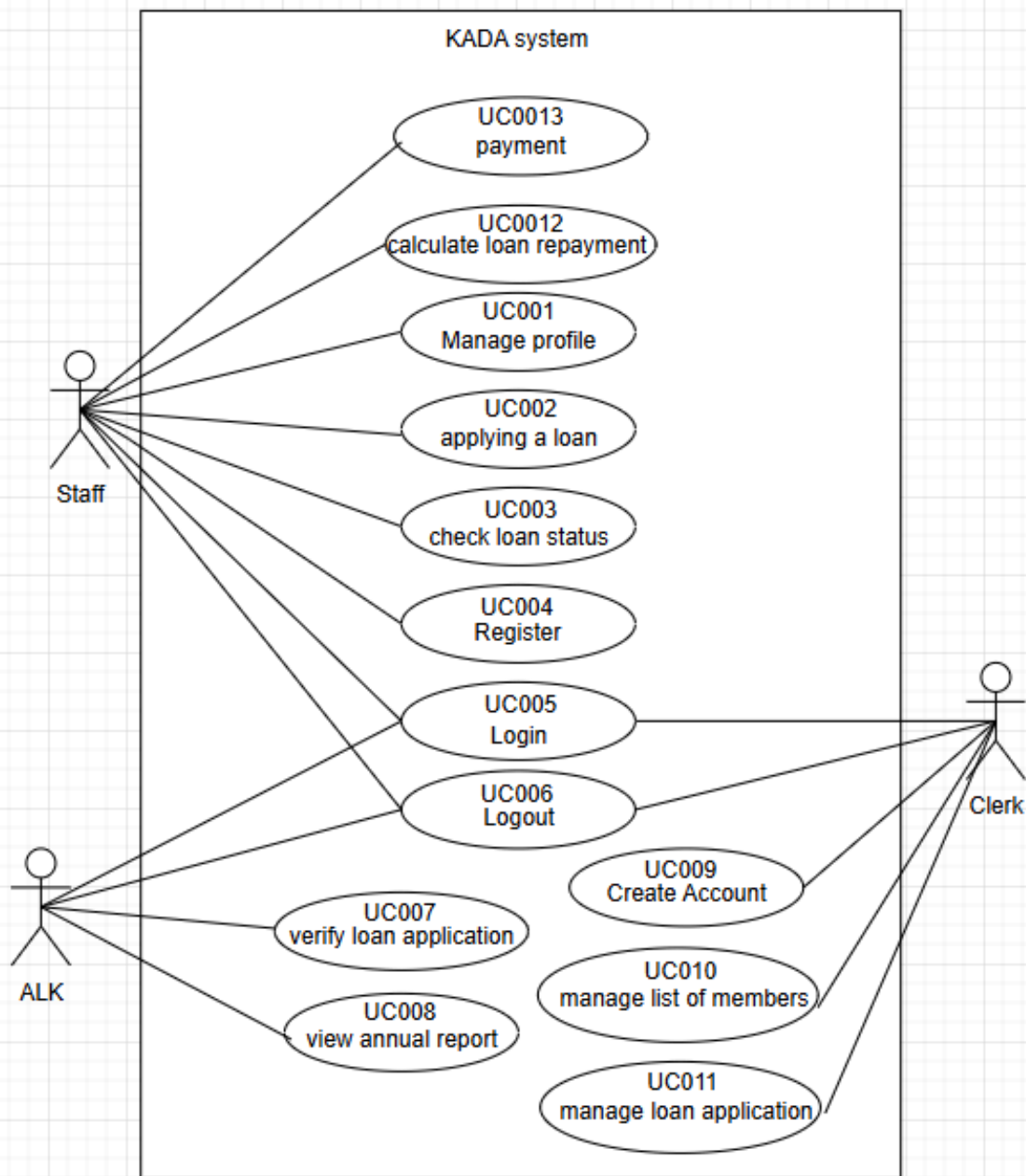


Loan Application Use Case Diagram



ALK and Payment usecase

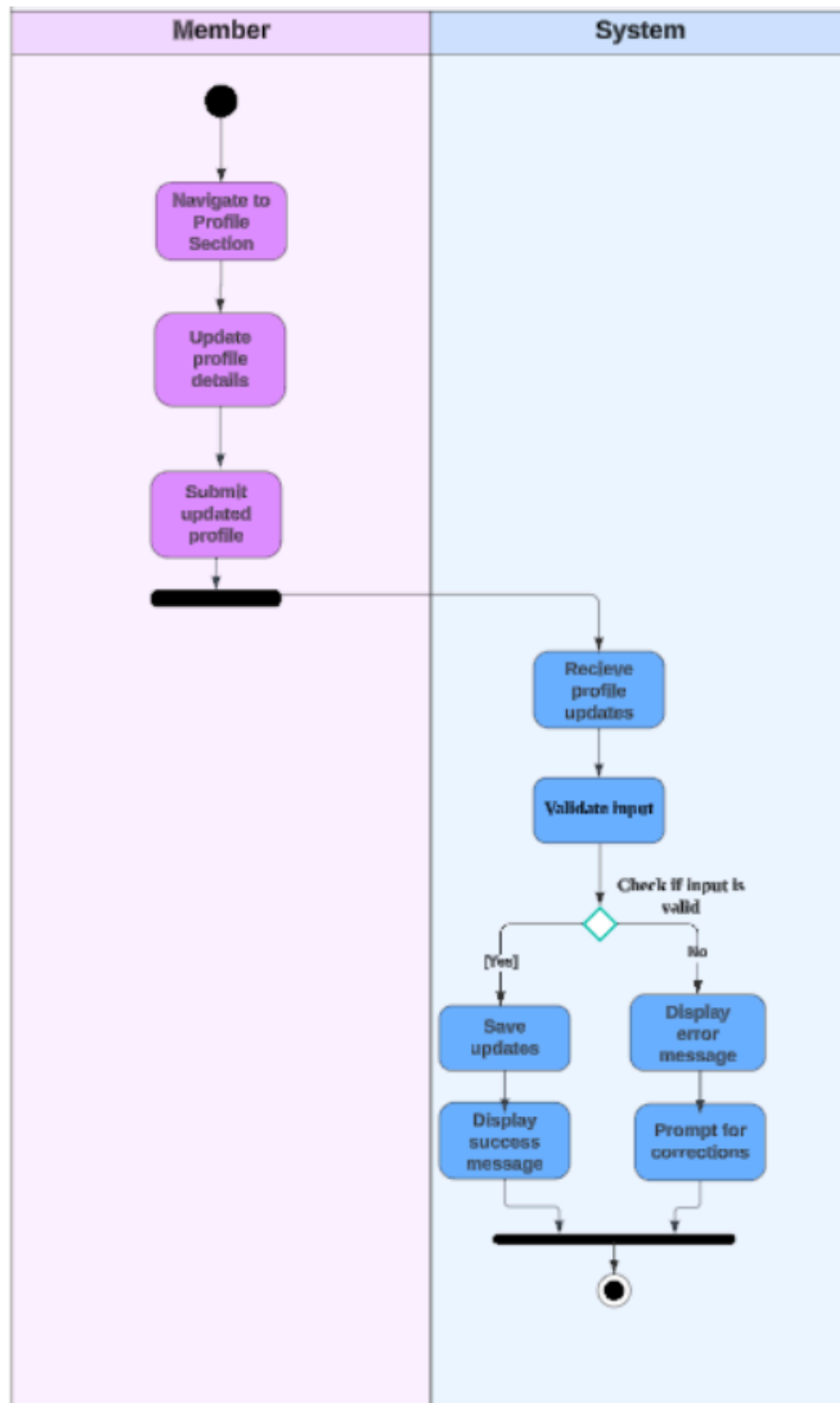




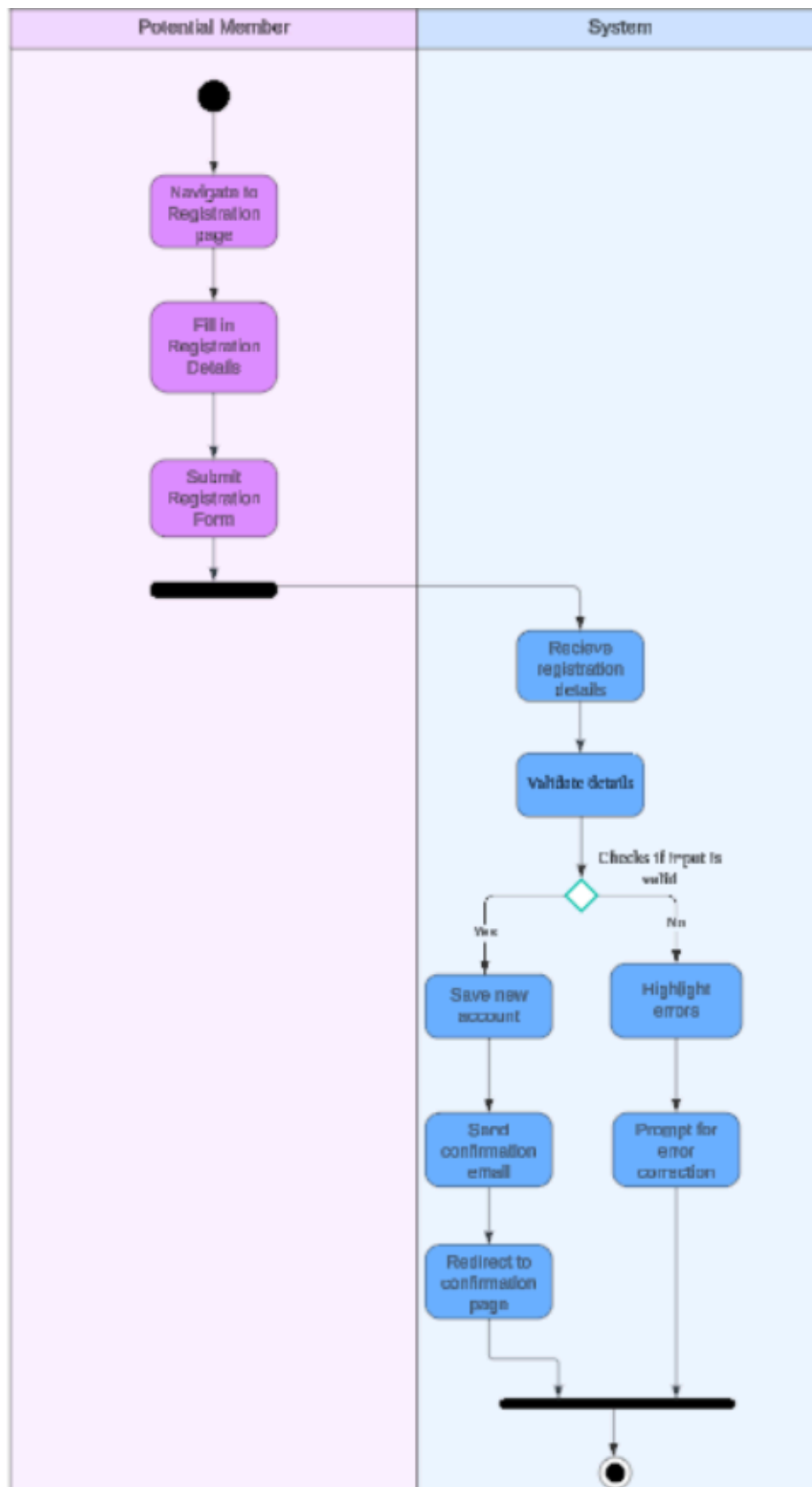
4.2 Activity Diagrams for every use case identified.

4.2.1 Module <Authentication>

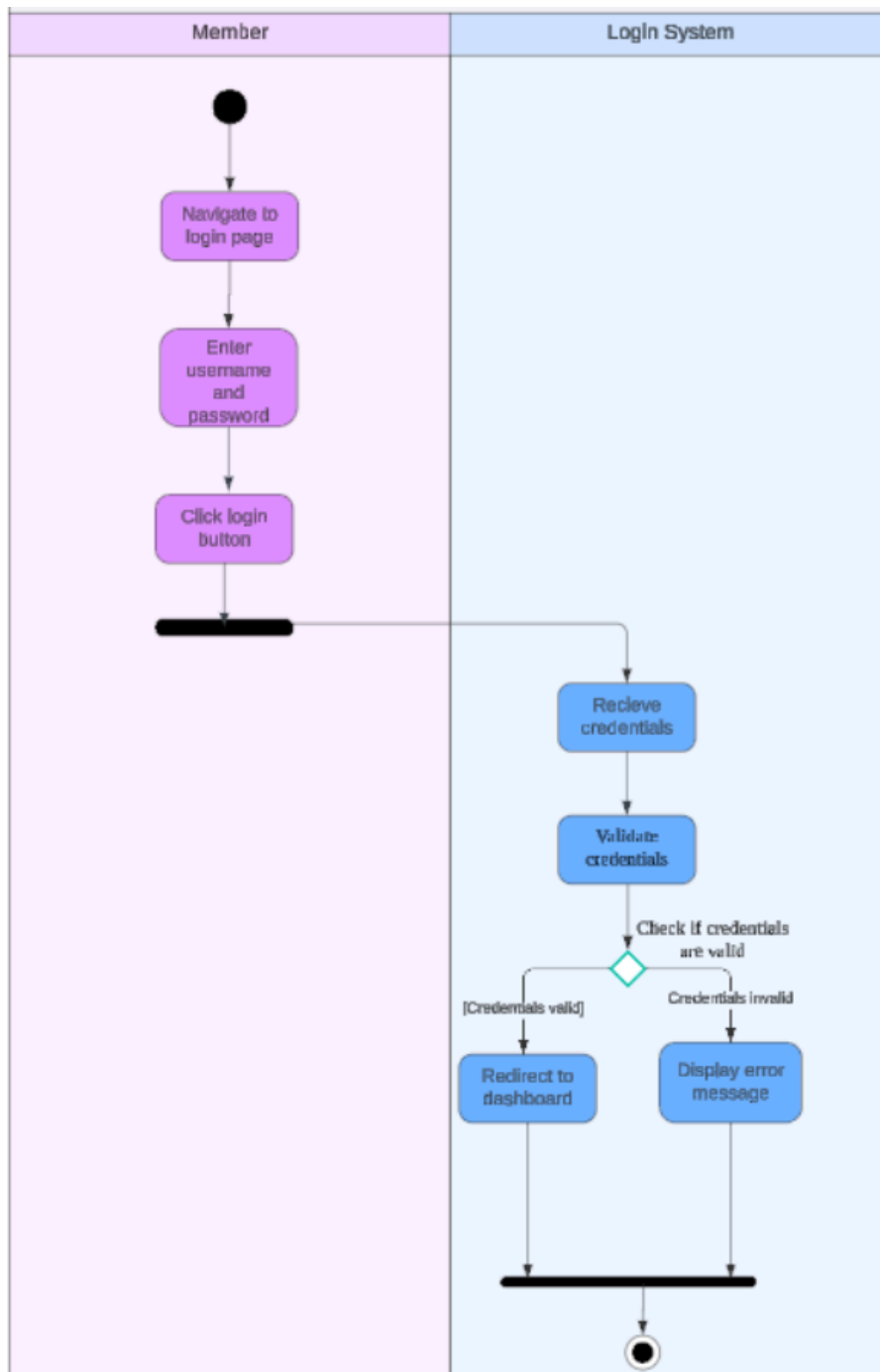
UC001 - Manage Profile



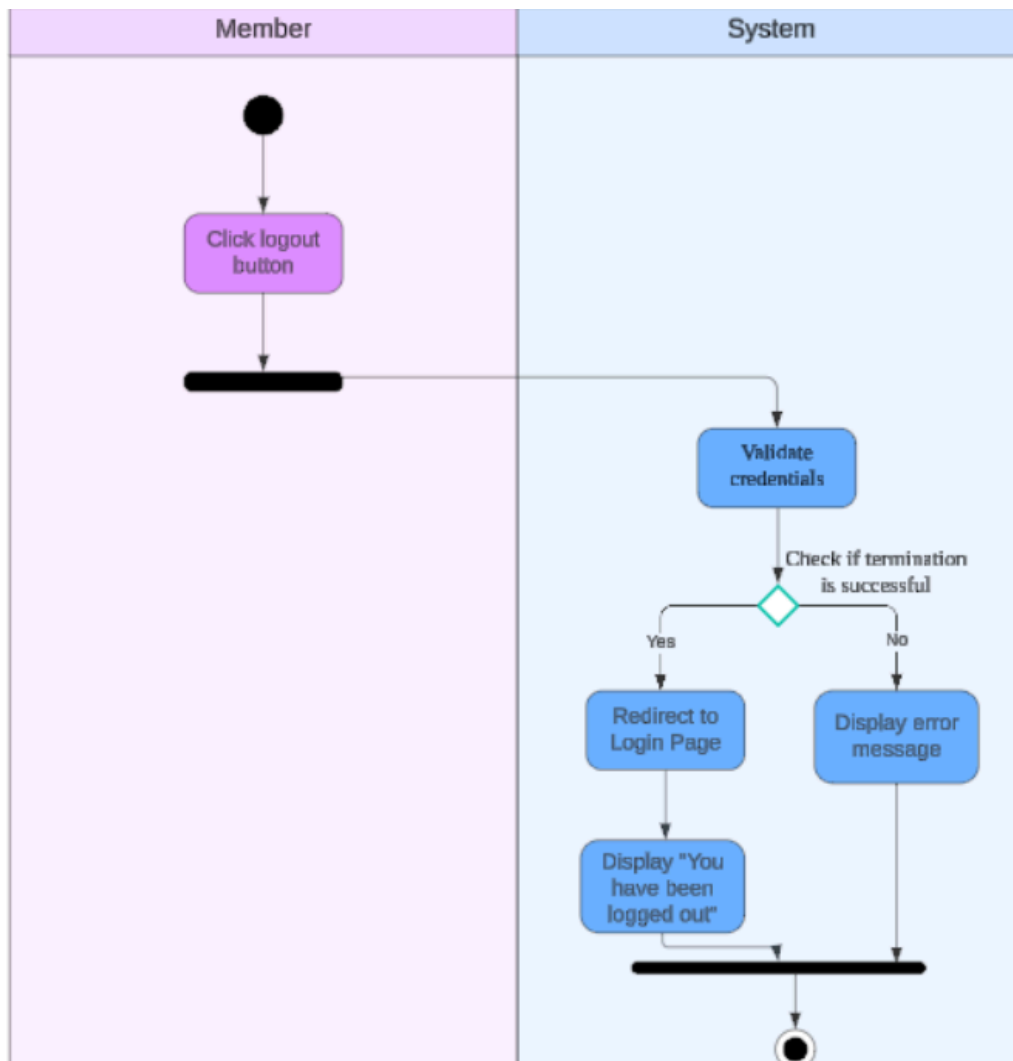
UC004 - Register



UC005 - Login

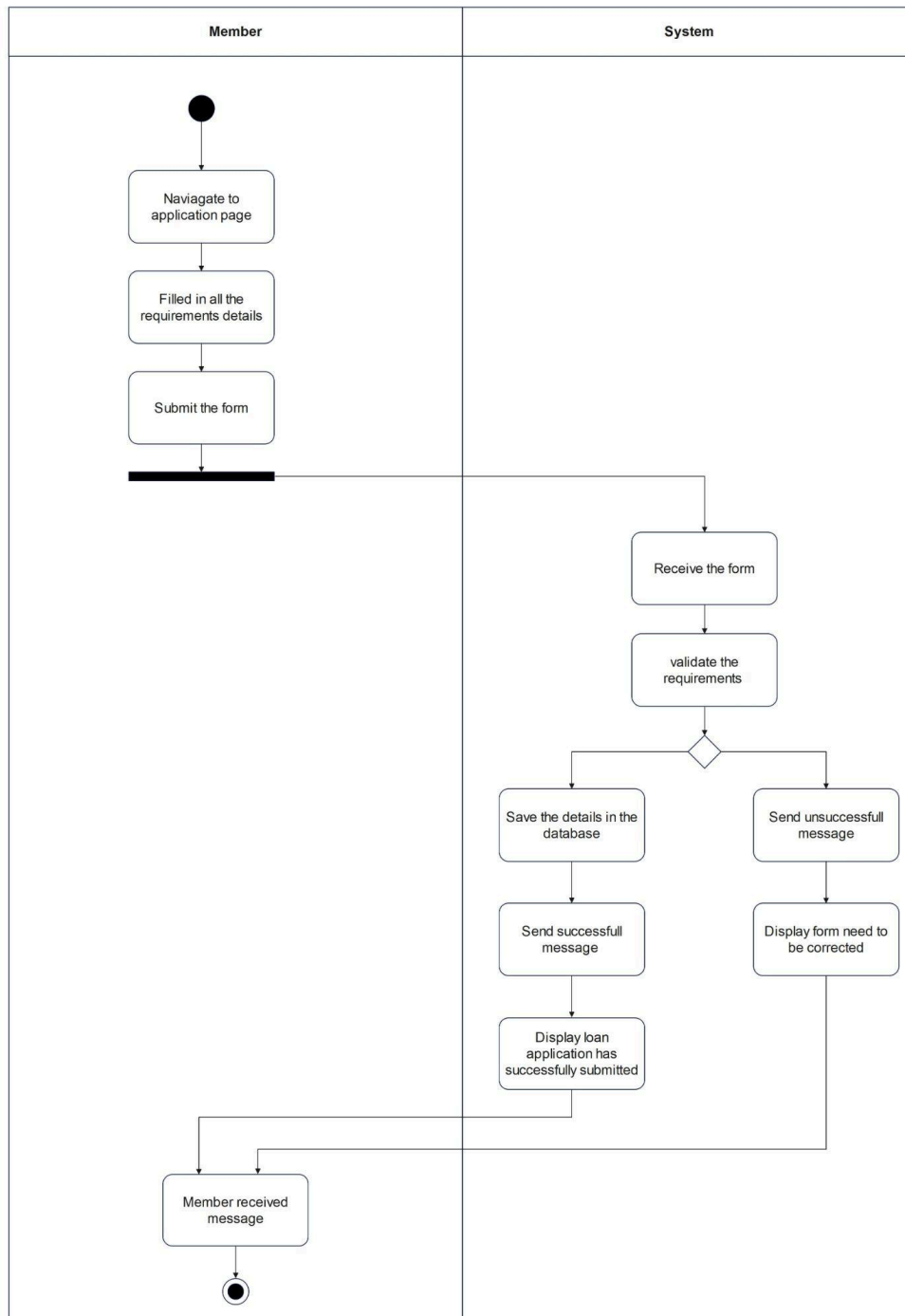


UC001 - Logout

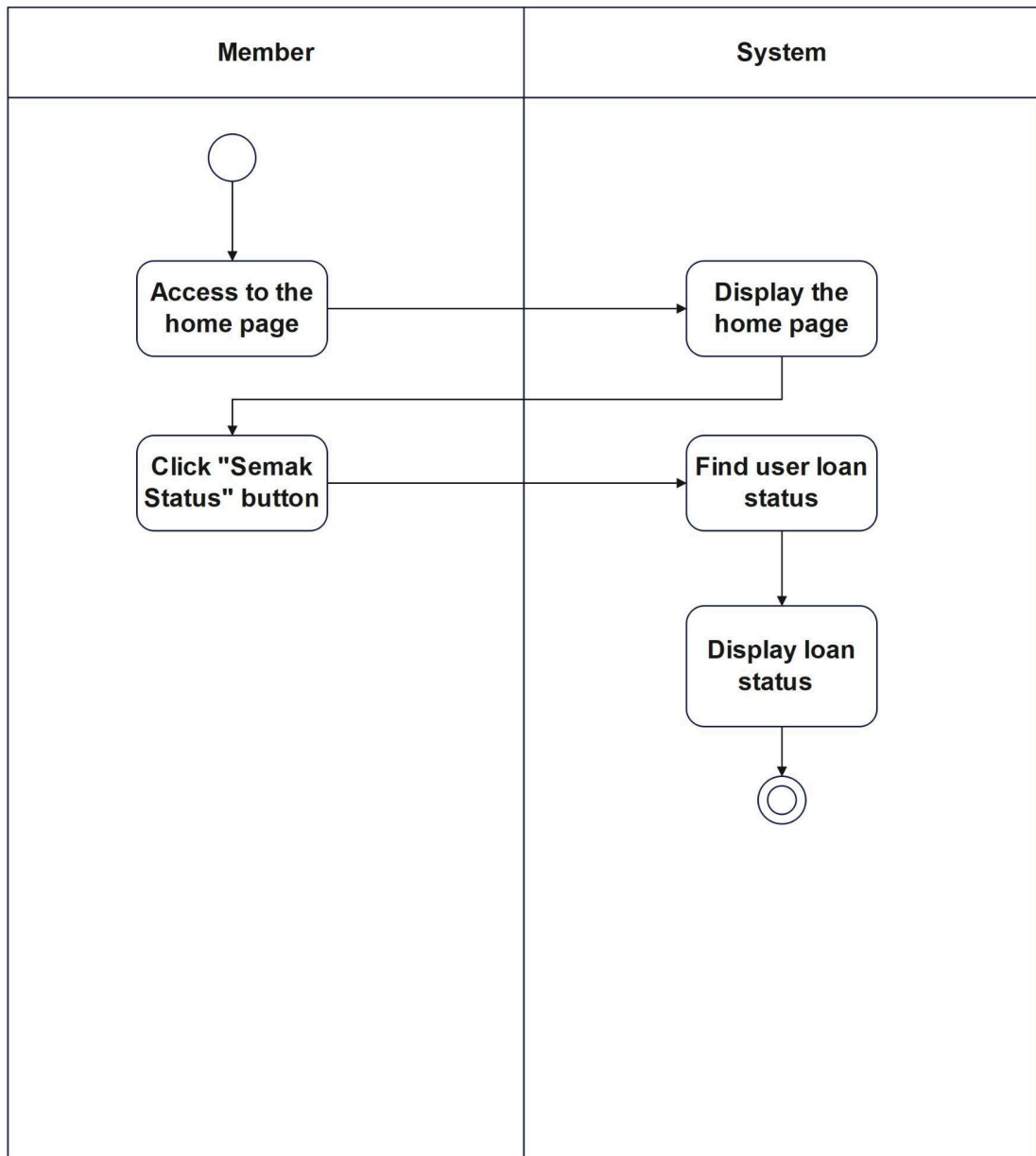


4.2.2 Module <Loan application>

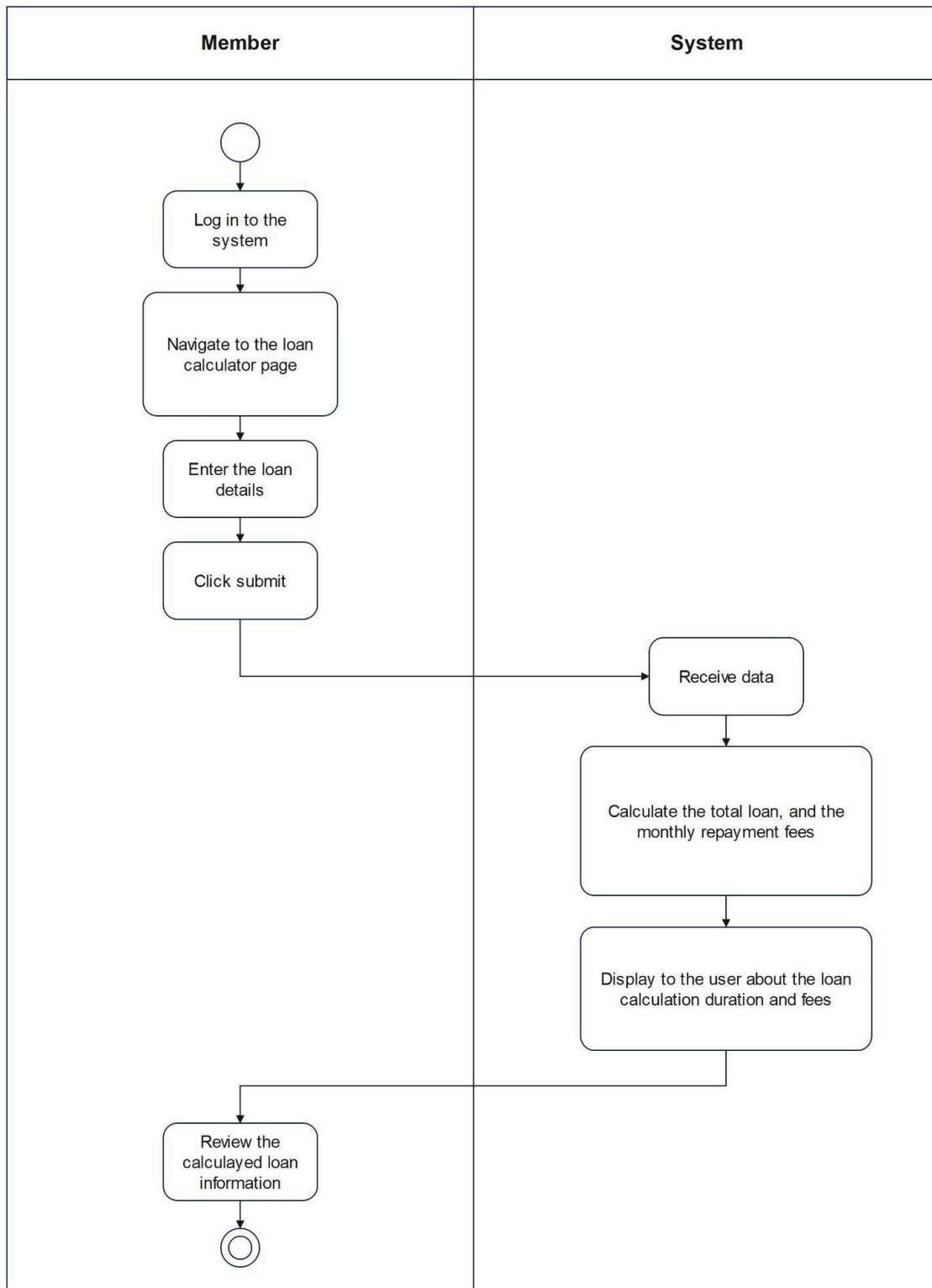
UC002 - Apply Loan



UC003 - check loan status



UC012 - Loan calculation



UC013 - loan payment

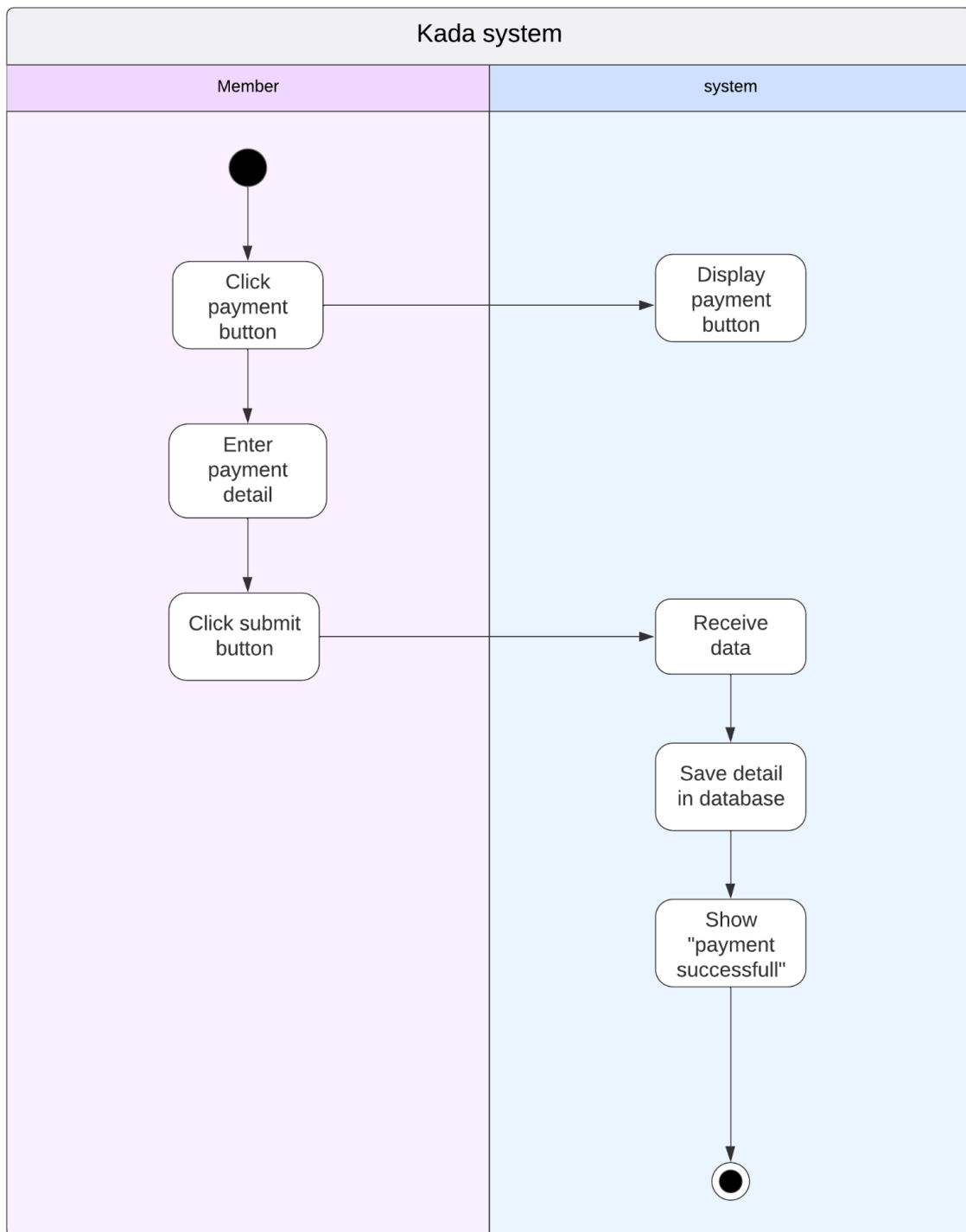


Figure 4.2.1.3: Activity Diagram for <<loan payment>>

4.2.3 Module <Clerk>

UC009 - Creating a Member Account

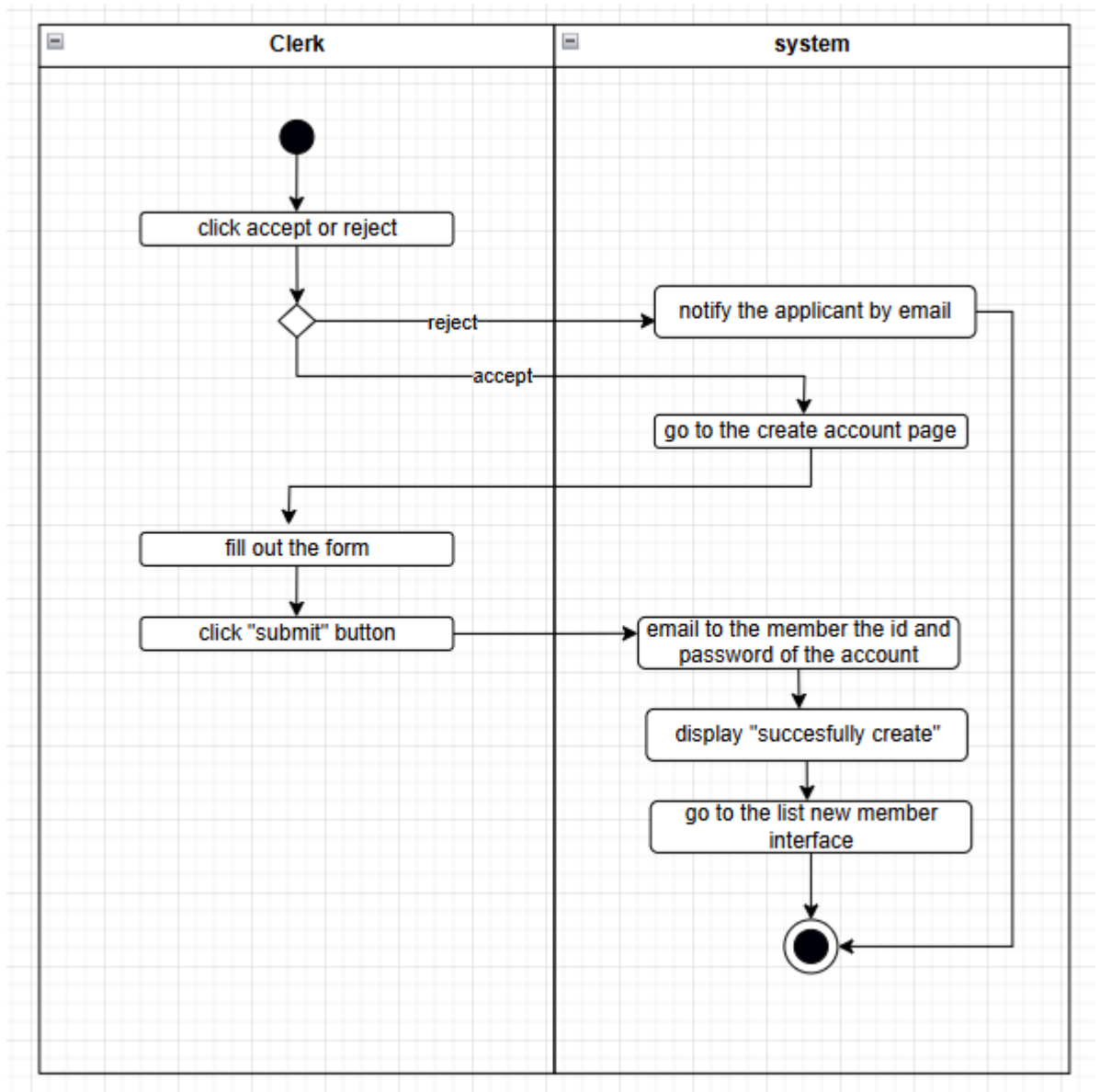
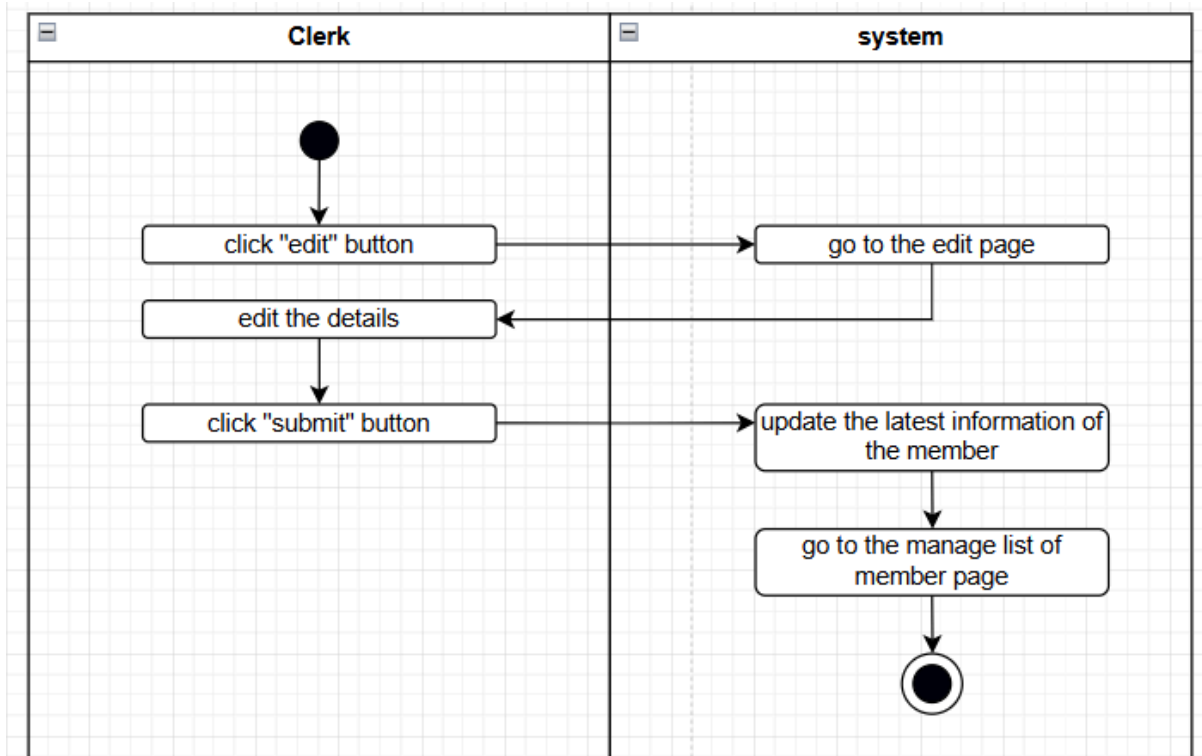


Figure 4.2.3.1: Activity Diagram for <<Manage Profile>>

UC010 - Managing Member Information



- Figure 4.2.3.2: Activity Diagram for <manage list of member (edit)>

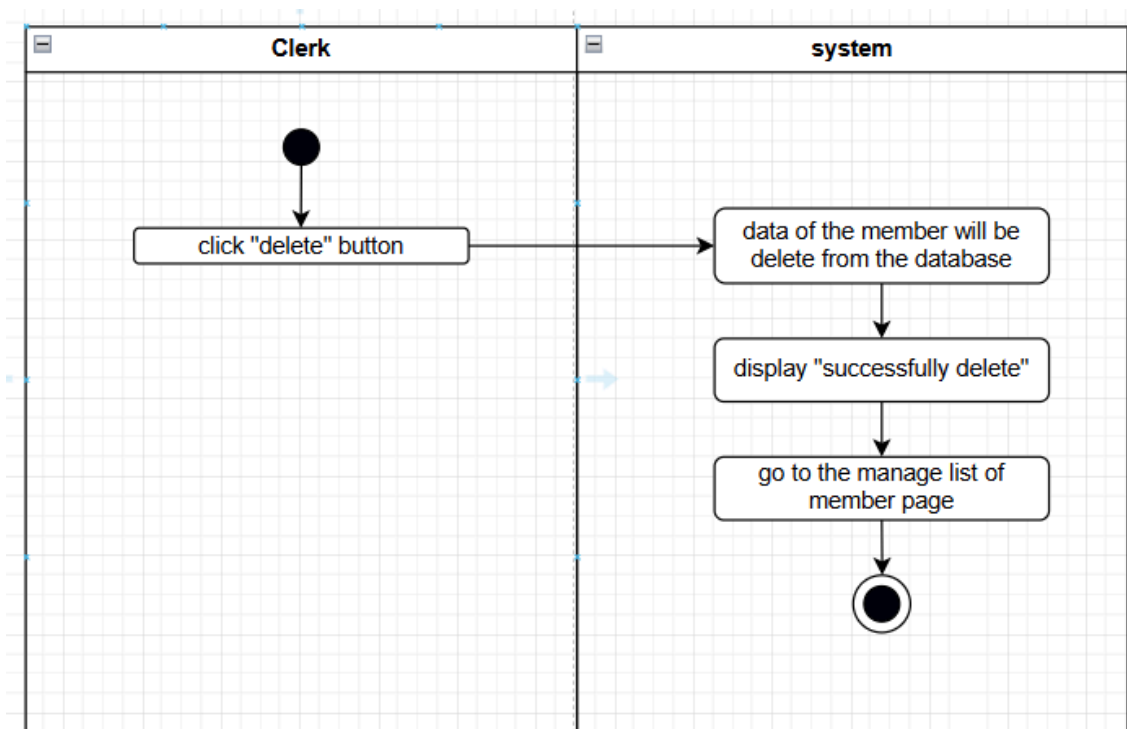


Figure 4.2.3.3: Activity Diagram for <manage list of member (delete)>

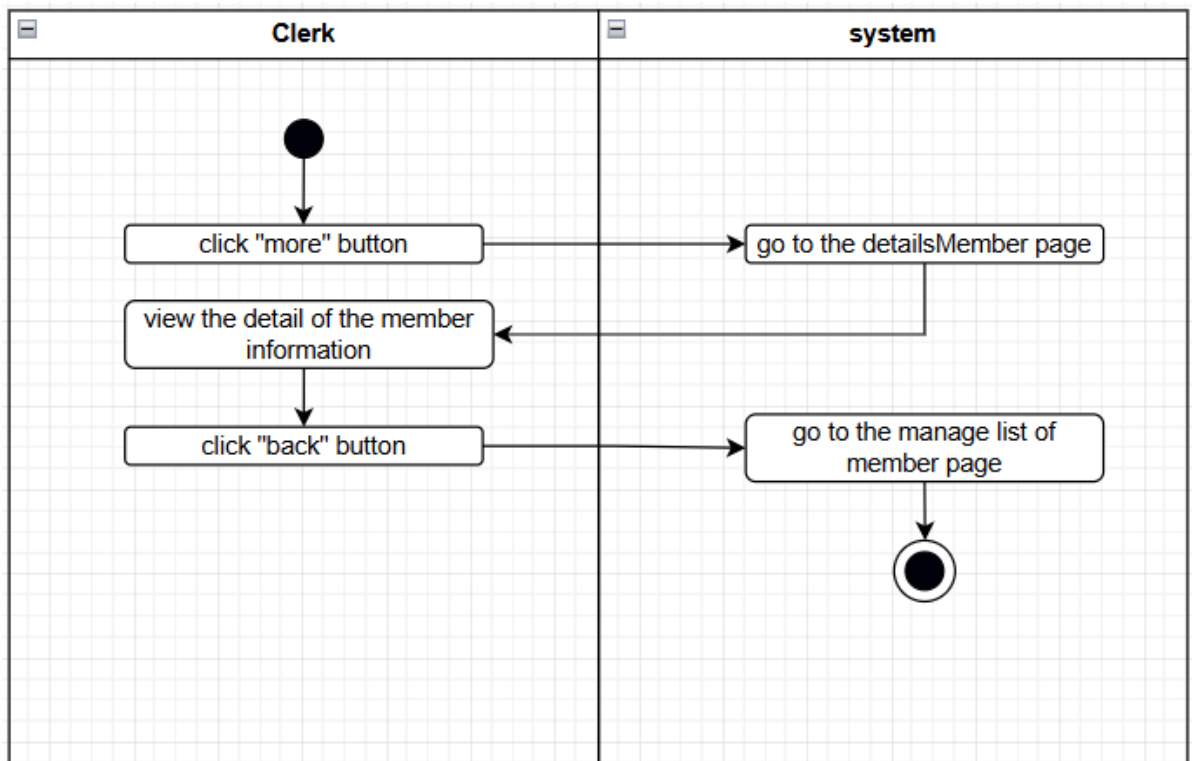


Figure 4.2.3.4: Activity Diagram for <manage list of member (more)>

UC011 - Managing Loan Application

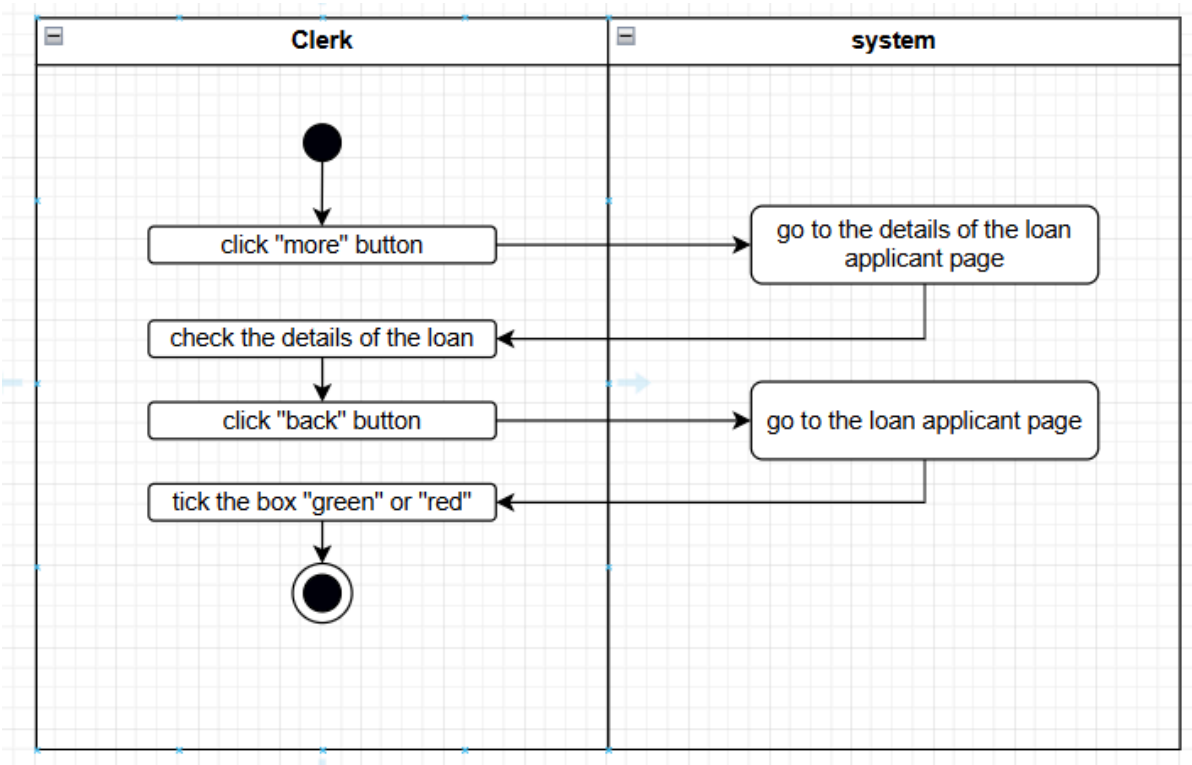


Figure 4.2.3.5: Activity Diagram for <manage list of member (view details)>

4.2.4 Module <ALK>

UC007 - verify loan application

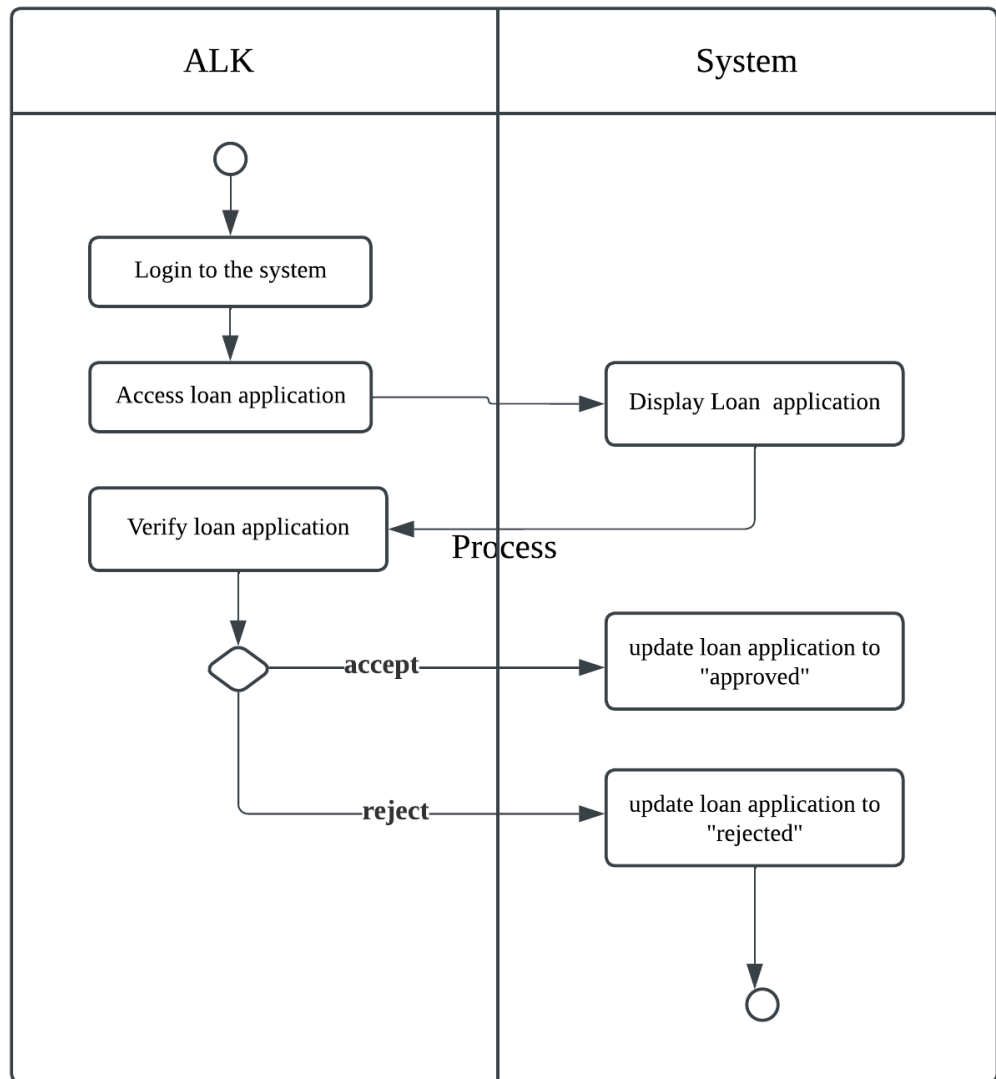


Figure 4.2.4.1: Activity Diagram for <<verify loan application>>

UC008 -view report

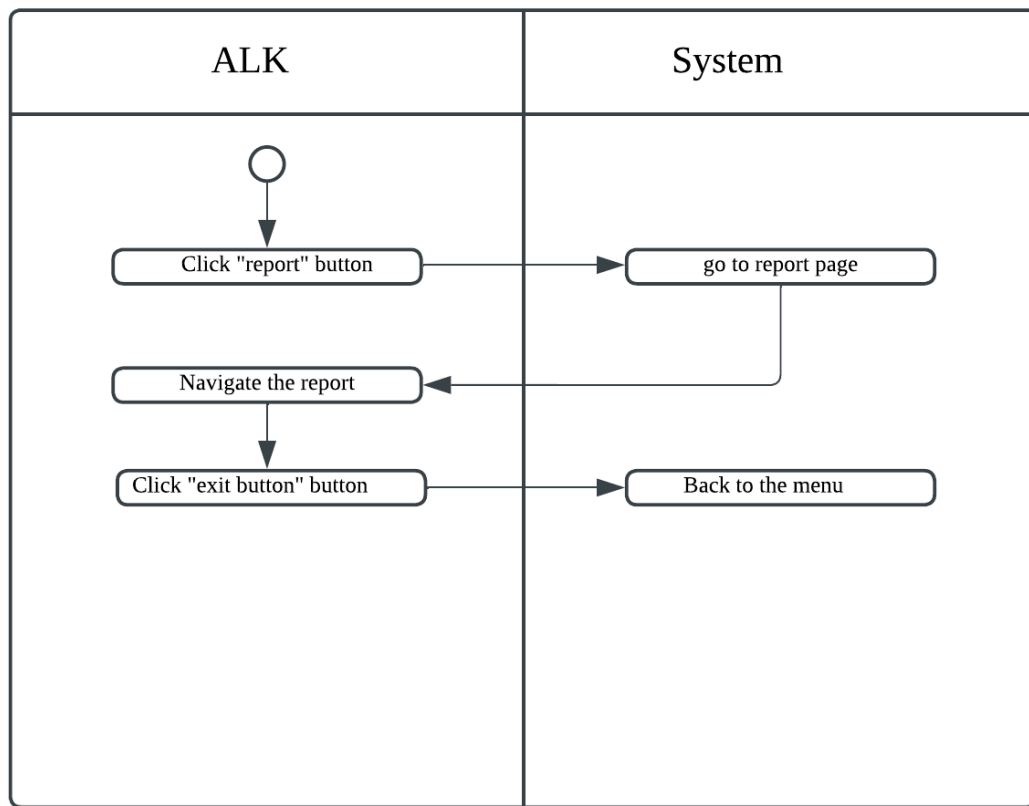


Figure 4.2.4.2: Activity Diagram for <<view report>>